ORLANDO I. ZARRÁGA MEDINA

Backend Developer | Node.js · NestJS · TypeScript · PostgreSQL
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PROFESSIONAL SUMMARY

Backend Developer specialized in TypeScript, NestJS and PostgreSQL, with experience designing APIs and SaaS platforms and taking them to production. Lead developer of ZarMedi and backend architect at InfinitySoftware; ZarMedi is a multi-tenant medical platform in production with 270+ endpoints, generative AI, a WhatsApp bot and digital signatures. Specialized in Domain-Driven Design (DDD), hexagonal architecture, security (JWT, RBAC, OAuth2), asynchronous processing with Redis and BullMQ, third-party API integrations and cloud deployment. I work to understand business goals and translate them into technical solutions with real impact, with clear communication and a strong commitment to quality.

PROFESSIONAL EXPERIENCE

ZARMEDI **2026 – Present**
Lead Developer *zarmedi.com*
Multi-tenant medical management SaaS in production for clinics and medical practices in Argentina and LATAM: scheduling, electronic health records, telemedicine, prescriptions, digital consent forms and WhatsApp automation. pnpm monorepo with 3 applications and 376 commits in 3 months.

- Architected the NestJS backend as a modular monolith with hexagonal architecture and DDD: 47 modules, ~272 versioned REST endpoints documented with Swagger, 41 entities and 109 TypeORM migrations on PostgreSQL (Supabase).
- Implemented subdomain-based multi-tenancy, multi-branch support and role-based access control (RBAC) with 6 roles and granular per-module, per-action permissions, plus a change audit log for full traceability.
- Built authentication with JWT and refresh tokens in HttpOnly cookies, passwordless one-time-password (OTP) access for patients, rate limiting and HTTP security headers.
- Integrated generative AI (Groq) for ICD-10-coded diagnosis suggestions, treatments, clinical summaries and demand forecasting, with per-clinic usage limits.
- Built a WhatsApp bot on Meta Cloud API that books, cancels and checks appointments 24/7, with a Redis-backed state machine and HMAC-SHA256 webhook verification, plus asynchronous processing with BullMQ (16 queues and 12 scheduled jobs).
- Developed digital signing of consent forms with SHA-256 document hashing and evidence logging (IP, user agent and timestamp), compliant with Argentina's Digital Signature Law 25,506.
- Integrated telemedicine (Daily.co and Socket.IO), Google Calendar (OAuth2), MercadoPago and Stripe payments via webhooks, transactional email (Resend), signed-URL storage and PDF prescriptions with verification QR codes.
- Backed the backend with 1,500+ Jest unit tests and deployed it on Railway with Docker; the frontend is a Next.js 14 PWA that consumes the API.

INFINITYSOFTWARE — CANARIAS EQUIPAMIENTOS **May 2026 – Present**
Backend Architect & Developer (Freelance) — 3-developer team

- Designed the layered architecture of the NestJS backend for a credit sales and collections management platform, and created 25 of its 26 API modules (~146 of 161 endpoints) on PostgreSQL and TypeORM.
- Implemented JWT authentication with Passport, multi-company operation with per-company data isolation and access control for 5 roles, along with the collection route sheets, daily closings, settlements, cash register and PDF/Excel reporting modules.
- Set up the pnpm monorepo, Docker containers and the deployment of the backend, frontend and database on Render.

MONETAE — EL SALVADOR **Jul 2026 – Oct 2026**
Full Stack Developer — Internship (3 months)

Compliance intranet (Compliance / HRIS)

- Developed the company's compliance intranet end to end with a NestJS backend following Domain-Driven Design (DDD), alongside the tech lead, and integrated the Next.js frontend with the API.
- Designed and implemented the identity and user management endpoints, integrated with Auth0 authentication.
- Established the team's GitHub workflow (protected branch, Pull Requests and two-reviewer approval) and onboarded two developers, delegating tasks and performing code reviews.
- Managed the deployment of the backend and database on Railway, coordinated with the frontend deployment on Netlify.

New company project

- Carried out the initial technical analysis, documentation, backlog and API contract ahead of development, and designed agents, subagents and micro-tasks for AI-assisted code generation with Claude Code.

MMI INTERNACIONAL · EQUIPAMIENTOS CANARIAS — SPAIN **Mar 2026 – Jun 2026**
Full Stack Developer — Internship (part time)

- Developed in Python an automated news content validator that decides publication through a scoring system, integrated with the company's internal publishing API.
- Integrated a local LLM (Ollama, Llama 3.2) to evaluate and correct headlines automatically, with a rule-based fallback using Jaccard similarity when the model is unavailable.
- Designed a resilient pipeline on Google Cloud Storage and AWS S3 with retries, automatic token renewal and idempotency control against duplicate publications, deployed as an AWS Lambda in Docker; contributed ~87% of the code, backed by 124 pytest tests and a 102-segment benchmark processed without errors.
- Refactored PHP code of the news management platform and contributed to the redesign of its interfaces.

TRAMA — PERU **Mar 2026 – Jun 2026**
Backend Developer — Internship (part time)

- As lead backend developer of TRAMA, a business-diagnostics SaaS for entrepreneurs, built 7 of its 10 modules and ~80% of the endpoints with NestJS, TypeScript, PostgreSQL and TypeORM, and integrated the team's Pull Requests.
- Implemented full authentication (access and refresh JWTs, hashed refresh tokens in httpOnly cookies, email verification and Google OAuth 2.0) and an admin panel with subscription-plan-based access control.
- Developed the TRAMA Index diagnostics engine with a weighted scoring algorithm, an event-driven achievements system and the deployment on Render with managed PostgreSQL.

IT RESOURCES INNOVATIONS LLC **May 2023 – Dec 2025**
Full Stack Developer (Freelance)

- Designed and developed microservices and REST APIs with NestJS and TypeScript following Clean Architecture, with JWT authentication and custom guards, integrating third-party APIs and managing versions with GitFlow.
- Optimized complex queries and transactions in SQL Server and MySQL, reducing response times by up to 30%.

ACADEMIC PROJECTS

¡SI CHEF! — HENRY BOOTCAMP **Nov 2025 – Jan 2026**
Backend / Full Stack Developer

- Developed the backend of a culinary social network with NestJS, PostgreSQL and TypeORM using a domain-based modular architecture, with JWT and OAuth authentication, role-based access control, Stripe payments, real-time features with Socket.IO and Google Cloud Vision and Vertex AI services.

EDUCATION

HENRY BOOTCAMP **2026**
Full Stack Web Development — 800 hours (graduated)

UNIVERSIDAD TECNOLÓGICA NACIONAL (UTN) **2024 – 2025**
Associate Degree in Programming — partial studies (30%)

CODO A CODO 4.0 **2021**
Full Stack Web Development (graduated)

TECHNICAL SKILLS & LANGUAGES

Languages: TypeScript, JavaScript (ES6+), Python, SQL, PHP

Backend: Node.js, NestJS, Express, REST APIs, Swagger/OpenAPI, WebSockets (Socket.IO), BullMQ

Architecture: Domain-Driven Design (DDD), hexagonal architecture, Clean Architecture, modular monolith, multi-tenancy, microservices

Security: JWT, OAuth2, Auth0, Passport, RBAC, OTP, HMAC, bcrypt, rate limiting

Databases: PostgreSQL, Supabase, Redis, MySQL, SQL Server, MongoDB, TypeORM, Prisma, Sequelize

Integrations: Generative AI (Groq, Ollama/Llama), WhatsApp Cloud API, Daily.co, Google Calendar, MercadoPago, Stripe, Resend, Cloudinary

DevOps & Cloud: Docker, AWS (Lambda, S3), Google Cloud Storage, Railway, Netlify, Vercel, Render, CI/CD, Git, GitHub, GitFlow, pnpm workspaces

Testing & Quality: Jest, Supertest, pytest, Pydantic, ESLint, code review

Frontend (complementary): React, Next.js, Vite, Tailwind CSS, Zustand, Zod

AI & Methodologies: Claude Code, AI agent-assisted development, Scrum, Kanban, backlog management

Languages (spoken): Spanish (native), English (intermediate B1 — technical reading and professional communication)